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ABSTRACT

Swordfish (*Xiphias gladius*) is a common target species of surface pelagic longline fisheries. In the Atlantic Ocean and Mediterranean Sea, swordfish is managed as three separate stocks, all having management measures in place to rebuild or conserve the stocks, including minimum landing sizes. The objective of this study was to review size data for swordfish in the Atlantic Ocean, model the sex-specific size distribution and determine areas where there is higher likelihood of capturing undersized fish. The size distribution differed between males and females and varied by quarter, indicating movements of large fish between temperate and tropical waters. Undersized fish seems to occur in association with coastal waters, with higher proportions in the Northwest Atlantic and tropical areas. This study provides a better understanding of the temporal and spatial size and sex distribution of swordfish and presents insights into the distribution of undersized swordfish that is subject to management measures.

KEYWORDS

Highly migratory species; lengths; minimum landing size; modeling; fisheries management

Introduction

Swordfish (*Xiphias gladius* Linnaeus, 1758) is the only species of the genus *Xiphias* and of the family Xiphiidae. It is a cosmopolitan species, widely distributed across all oceans, in tropical, temperate, and cold waters (Nakamura 1985). Swordfish are mostly captured in pelagic longlines, being a target species for surface pelagic longlines and a bycatch species in deeper pelagic longlines targeting tunas (Ward et al. 2000; Domingo et al. 2014). Global catches of swordfish in 2021 reached 96,410 tonnes (FAO, 2023). The International Commission for the Conservation of Atlantic Tunas (ICCAT), responsible for managing tuna and tuna like species in the Atlantic Ocean and adjacent Seas, reported in 2021 a total catch of 26,727 tonnes in the Atlantic and Mediterranean Sea (ICCAT, 2023a), which represents around 30% of the global catches.

In the ICCAT area, swordfish is managed as three separate stocks; two in the Atlantic Ocean, separated at the 5°N latitude, and one in the Mediterranean Sea (Abid and Idrissi 2006); although there are movements of individuals between them (Smith et al. 2015; Rubio et al. 2022). In the late 1980s and 90s, the stocks were subject to overfishing and became overfished in subsequent years in terms of Maximum Sustainable

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Yield (MSY) reference points (ICCAT, 2023b). Several management measures were adopted for each of the stocks to rebuild or maintain the stocks at sustainable levels. For the Atlantic stocks, measures have been in place since the early 1990s to rebuild them, including total allowable catch (TAC) limits, and minimum retention size limits (Neilson et al. 2013). In the Mediterranean Sea, measures have been implemented since the early 2000s, including a TAC, minimum retention sizes and time closures (ICCAT, 2016). The North Atlantic stock has been considered recovered since 2009, which has been attributed to the implemented measures, but also to the resilience of the stock and good recruitment in the late 1990s to early 2000s (Neilson et al. 2013). Additionally, reported catches have remained below the TAC for most years since implementation of the management measures (ICCAT 2023b), which appears to have contributed to the recovery of the stock. The most recent stock assessment of the Northern stock corroborates that this stock is rebuilt and is being maintained at levels consistent with MSY, with the status of the stock assessed as not overfished, and not undergoing overfishing (ICCAT, 2023b). In the South Atlantic, the stock is still overfished, and overfishing is occurring, while in the Mediterranean Sea, the stock is overfished but overfishing is not occurring (ICCAT, 2023b).

Given the swordfish importance for pelagic fisheries and its exploitation status, Atlantic and Mediterranean swordfish have been subject to several studies regarding age and growth (e.g., Tserpes and Tsimenides 1995, Arocha et al. 2003, Garcia et al. 2017); reproduction (e.g. Arocha and Lee 1996; De la Serna, 1996; Hazin et al. 2002), sex ratios (e.g. Abid et al. 2018; Mejuto 2018), movements (e.g. Neilson et al. 2009; Neilson et al. 2014; Abascal et al. 2015; Braun et al. 2019) and distribution of catch rates and habitat preferences (e.g. Domingo et al. 2007; Hazin and Erzini 2008; Chang et al. 2012; Chang et al. 2013; Schirripa et al. 2021; Nóbrega et al. 2023; Lima et al. 2023; Parra et al. 2023). Some size distribution studies exist in specific areas as in the equatorial South Atlantic Ocean (Su et al. 2021), Caribbean Sea and adjacent waters of the Western Central Atlantic Ocean (Arocha et al. 2017) and the southwest Atlantic Ocean (Hazin and Erzini 2008), or in broader Atlantic areas (Mejuto 2003). Understanding the species biology and population dynamics, including its spatio-temporal distribution, is essential for fisheries management, as potential impacts by fisheries can be better identified, and the scientific advice provided becomes more specific (Coelho et al. 2018).

To date, an oceanic-wide and fleet-combined study of the variation in the size distribution of swordfish at an ocean-wide scale is lacking. Consequently, the main goal of this study is to review and model the detailed size distribution data available for swordfish from the major oceanic longline fleets that target swordfish and/or tunas in the Atlantic Ocean with a view to: 1) describe and summarize the seasonal variation in the sex and size distribution of swordfish in the Atlantic Ocean; and 2) identify the main areas of occurrence of undersized fish and their expected probability of occurrence.

Data review and description

Swordfish records and data were collected by national scientific observers onboard commercial vessels, detailed logbooks data, port sampling from national data collection programmes, and scientific surveys, from 1979 to 2022 (Table 1). For captured specimens, data on size, sex, capture location and date were recorded. The size measurement used was straight lower jaw fork length (LJFL). When other length measurements were provided (e.g., curved lower jaw fork length, eye fork length) these were converted to LJFL using the size-size conversion equations in Rey & Gonzales-Garces (1979) and Coelho et al. (2022a). In cases where weight was recorded, with no length record available, these were converted to LJFL using equations provided by Arocha (1997). Records below 30 cm LJFL and above 1000 kg (442 cm LJFL) were considered unrealistic and were excluded from the analysis.

This resulted in 452,336 swordfish records from a wide geographical range in the Atlantic Ocean (Table 1), including the Mediterranean Sea (longitude <5°E), and waters of the Indian Ocean adjacent to the South Atlantic Ocean around the Cape of Good Hope (longitude <40°E) (Figure 1). Specifically, data were obtained between 56°N to 45°S latitude and spanned the entire longitudinal range of the Atlantic area, as well as the adjacent seas, with 270,482 records from the North Atlantic (lat > 5° N), 110,055 from the South Atlantic (lat < 5°N), 67,491 from the Mediterranean Sea (between 5°W and 5°E in the North Atlantic), and 4,308 from the Indian Ocean adjacent to the Atlantic (between 20° and 40°E in the South Atlantic Ocean).

Most data originated from commercial drifting pelagic longlines (91.9%), but harpoon (6.7%) and gillnet (1.4%) data were also provided, as well as a few records from handline were also included (<0.01%). Specimens included in the analysis ranged in size from 30 to 411 cm LJFL, covering the known size range of the species. Sex was recorded for 227,582 specimens (Table 1), including 3,361 from the Indian Ocean and 278 from the

Table 1. Summary of the compiled size data for swordfish in the Atlantic Ocean and Mediterranean Sea, by flag, gear and source. Information on the sample size in number of specimens (N) by sex (female, male and unknown), size range and mean size (LJFL—lower jaw fork length, cm) and the range of years.

					N			Size (LJFL cm)		Year
	Flag	Stock	Gear	Source	Female	Male	Unk.	Range	Mean	Range
Atlantic	Brazil	Both	Longline	Observer	5394	7152	16404	50 - 367	152.0	2005-2022
	Canada	North	Hand line[1]	Observer	3	1	2	119 - 199	150.5	2009-2021
		North	Harpoon[1]	Observer	4	3		172 - 235	195.6	2003-2009
		North	Harpoon[2]	Port sampling			30084	39 - 350	204.4	1999-2022
		North	Longline	Observer	22487	11536	5764	58 - 411	157.5	1979-2022
	Chinese Taipei	Both	Longline	Observer	2609	2973	1047	47 - 295	161.2	2015-2022
	Côte D'Ivoire	South	Gillnet[2]	Port sampling			2303	33 - 268	158.3	2013-2015
	Portugal	Both	Longline	Observer	9230	11897	23298	45 - 312	140.0	2003-2022
	Spain	Both	Longline	Electronic	325	224	4204	50 - 361	141.5	2019-2022
				Observer	6803	6954	15569	35 - 305	143.5	2017-2022
		North	Longline	Observer	97	77	8963	34 - 305	135.5	2009-2022
	Japan	North	Longline	Observer	5264	2687	2164	35 - 337	150.7	1979-2000
		Both	Longline	Observer	4627	2043	1846	38 - 278	143.4	1997-2020
	South Africa	South	Longline	Observer	2497	1429	2066	64 - 314	166.9	2012-2022
	Uruguay	South	Longline	Observer	15134	7463	8724	40 - 322	156.3	1998-2013
				Research Campaign	165	91	168	72 - 306	154.3	2009-2019
	USA	Both	Longline	Observer	48224	39189	32144	30 - 293	131.0	1992-2022
	Venezuela	North	Gillnet	Port sampling	1299	714	2146	67 - 257	128.0	1991-2014
		Both	Longline	Observer	4578	4131	645	41 - 293	131.3	1987-2018
Mediterranean	Spain		Longline	Observer	173	105	67213	30 - 255	110.2	2009-2022

[1]Used only in descriptive analysis; [2]Used only in proportion of undersized model.

Figure 1. Location of the collection of swordfish (*Xiphias gladius*) size samples for this study in the Atlantic Ocean and Mediterranean sea. Size classes in lower jaw fork length (LJFL, cm) are represented by colored circles. The categorization of size-classes was carried out using the 0.2 quantiles of the data (values in the legend represent the lower and upper limits of each sizeclass). The eleven areas defined for this study are identified with dashed lines. CAN – waters offshore to Canada; GOM – Gulf of Mexico; NNW - north northwest; NNE - north northeast; NSW - north southwest; NSE - north southeast; MED – Mediterranean; SNW - south northwest; SNE - south northeast; SSW - south southwest; SSE - south southeast.

Mediterranean Sea. In total, 128,913 individuals (56.6%) were females and 98,669 (43.4%) were males, representing an overall female to male sex ratio of 1.31:1.

Considering the distribution of sizes of swordfish in the sample and within each stock area definition, the Atlantic Ocean was divided into eleven areas. In the North

Atlantic (latitude >5°N), six areas were considered: the northwest (NNW), northeast (NNE), southwest (NSW), southeast (NSE), waters off Canada (CAN) and the Gulf of Mexico (GOM). In the South Atlantic (lat <5°N), four areas were defined: northwest (SNW), northeast (SNE), southwest (SSW), and southeast (SSE) which includes individuals captured in the Indian Ocean around the Cape of Good Hope, as this is a potential area of mixing between the Atlantic and Indian Oceans stocks. Hereafter, the Indian Ocean records will be reported as belonging to the South Atlantic, in area SSE. The Mediterranean Sea (MED) was represented by a single area.

The analysis for this study was carried out using the R language for statistical computing version 4.3.1 (R Core Team 2023). Specific functions and packages ae referred to when appropriate, additional packages used included the following: "car" (Fox and Weisberg 2019), "classInt" (Bivand 2023), "data.table" (Dowle and Srinivasan 2023), "dplyr" (Wickham et al. 2023), "geomtextpath" (Cameron and van den Brand 2022), "ggplot2" (Wickham 2016), "ggspatial" (Dunnington 2023), "nortest" (Gross and Ligges 2015), "patchwork" (Pedersen, 2023), "perm" (Fay and Shaw 2010), "plyr" (Wickham 2011), "raster" (Hijmans 2023), "stringr" (Wickham 2023), "viridis" (Garnier et al. 2023).

Size data were tested for normality with Kolmogorov–Smirnov normality tests with the Lilliefors correction (Lilliefors 1967), and for homogeneity of variances with Levene tests (Levene et al. 1960). Size data were not normally distributed (Lilliefors test: $D = 0.053$, $p < 0.001$), and the variances were heterogeneous among areas (Levene test: $F = 2205.5$, df = 10, $p < 0.001$), quarters (Levene test: $F = 1709.7$, df = 3, $p < 0.001$) and sexes (Levene test: $F = 8019.3$, df = 1, $p < 0.001$). Specimen sizes were compared among areas, sexes and quarters of the year (Q1: January – March, Q2: April – June, Q3: July – September, Q4: October – December) using non-parametric k-sample permutation tests (Manly 2007), which revealed that sizes differed significantly among areas (permutation test: chi- squared = 129150, df = 10, $p < 0.001$), quarters (permutation test: chi-squared = 8814.5, df = 3, $p < 0.001$) and sexes (permutation test: $Z = 114.54$, $p < 0.001$).

Swordfish size distribution in the Atlantic Ocean

Spatio-temporal size distribution

Size frequency distributions and trends in mean size were analyzed and plotted by area, sex and quarter of the year. In general, larger individuals were present in higher latitudes of both the North and South Atlantic Oceans (Figure 1). In the Atlantic Ocean, the larger individuals were found in the waters off Canada, while the smaller ones were captured in the Gulf of Mexico (GOM) (Figure 2). Compared to the Atlantic Ocean, Mediterranean Sea specimens were smaller and had the lowest mean size (Figure 2). Most areas presented a unimodal length distribution (Figure 2), except for the GOM that presented a marked bimodal distribution with a peak of individuals around 80 cm LJFL and another at around 125 LJFL cm. The NNW and MED presented small peaks at around 80 and 50 cm LJFL, respectively. Within the general size distribution across areas, males tended to be smaller than females. These differences were more evident in the areas with overall larger individuals (Figure 3).

Size trends in each area also changed with quarters (Figure 4). The areas with the most pronounced trends were CAN and MED. The MED presented a decrease in mean size from quarters 2 and 3 to quarters 4 and 1. In the CAN area, larger sizes were observed in quarters 2 and 3 while in quarters 4 and 1 there is a decrease in size. To a lesser extent area NNW presents a similar pattern to CAN, while areas NSW and NSE have larger individuals in quarters 1 and 2, while quarters 3 and 4 there is a decrease in mean size. In the South Atlantic Ocean, size trends throughout the year are less variable, with area SSE seeming to have larger mean sizes later in the year, quarters 3 and 4.

Predicting swordfish size distribution

Generalized Additive Mixed Models (GAMMs) were applied to identify the spatio-temporal size distribution of swordfish and proportions of undersized fish (below). Environmental variables derived from remote sensing (sea surface temperature) and static physiographic variables (bathymetry and distance to coastline) were included in the analysis. Daily sea surface temperature (SST; °C) data were obtained from the NOAA 0.25° Daily Optimum Interpolation Sea Surface Temperature (Huang et al. 2020), which starts in 1982 and is available up to the present, using the *xtracto* function from the *rerddapXtracto* package (Mendelssohn 2023a) in R language. Bathymetric values were obtained from the ETOPO1 Global Relief Model at a global 1 arc-minute grid (Amante and Eakins 2009), using the *xtractomatic* function from the *xtractomatic* package (Mendelssohn 2023b) in R language (R Core Team 2023). Minimum distances from each location to the coastline were calculated with the *dist2land* function of the ggOceanMaps package (Vihtakari 2023).

Figure 2. Size (lower jaw fork length, LJFL in cm) frequency distribution of swordfish (*Xiphias gladius*) caught in the different areas of the Atlantic Ocean and Mediterranean Sea from 1979 to 2022. Filling colors represent different stocks (dark blue for North Atlantic Ocean – 6 areas; black for the Mediterranean Sea (1 area); and light blue for the South Atlantic Ocean (4 areas)). The dashed black lines are drawn at the mean size for the stock (North Atlantic - 144.4 cm, South Atlantic - 154.4 cm and Mediterranean Sea - 110 cm LJFL), the solid orange line is drawn at 125 and 100 cm LJFL, for the Atlantic Ocean and Mediterranean Sea, respectively. CAN – waters offshore to Canada; GOM – Gulf of Mexico; NNW - north northwest; NNE - north northeast; NSW - north southwest; NSE - north southeast; MED – Mediterranean; SNW - south northwest; SNE - south northeast; SSW - south southwest; SSE - south southeast (see Figure 1).

Data on 226,779 individuals with assigned sex was modeled after exclusion of data before 1982 for which SST data was not available, additionally points on land were also excluded. The explanatory variables considered were latitude and longitude (continuous), month, year, sex, bathymetry, distance to coastline, sea surface temperature, and fleet (a combination of flag, gear and area of operation, see Table 1).

The GAMM model for size was expressed as follows:

$$g(\text{Size}) = \text{Sex} + \text{Month} + \text{Month}:\text{Sex} + f(\text{Latitude}:\text{Longitude}, \text{Month}, \text{Sex}) + f(\text{Year}) + f(\text{SST}) + f(\text{Bathymetry}) + f(\text{dtl}) + \alpha\text{Fleet}$$

The model used a gamma error distribution which is a continuous probability distribution with a shape and a scale parameter, and was fit with the log link function (g). Definitions for the explanatory variables were the following:

Sex as a categorical variable (male or female); Month as a categorical variable (January to December); Sex:Month as the interaction term between the two categorical variables; f(Latitude:Longitude, Month, Sex) as a two dimensional smoother estimated with a spline on the sphere for the spatial effect by month and sex; *f*(Year), *f*(SST), *f*(Bathymetry), *f*(dtl) as smoothers for the effect of the continuous variable year (1982-2022), sea surface temperature (SST), bathymetry and distance to coastline (dtl) estimated with a thin plate regression spline. αFleet as a random intercept effect

Figure 3. Size (lower jaw fork length, LJFL cm) frequency distribution of swordfish (*Xiphias gladius*) caught in the different areas of the Atlantic Ocean and Mediterranean by sex from 1979 to 2022. The horizontal orange line is drawn at 125 and 100 cm LJFL, for the Atlantic Ocean and Mediterranean Sea, respectively. Circles inside the violin plot represent the mean size by area and sex. CAN – waters offshore to Canada; GOM – Gulf of Mexico; NNW - north northwest; NNE - north northeast; NSW - north southwest; NSE - north southeast; MED – Mediterranean; SNW - south northwest; SNE - south northeast; SSW - south southwest; SSE - south southeast (see Figure 1); F – female; M – male.

Figure 4. Boxplot of swordfish (*Xiphias gladius*) sizes (lower jaw fork length, LJFL cm) in the Atlantic Ocean and Mediterranean Sea by quarter from 1979 to 2022. The dashed line is drawn at 125 and 100 cm LJFL, for the Atlantic Ocean and Mediterranean Sea, respectively. CAN – waters offshore to Canada; GOM – Gulf of Mexico; NNW - north northwest; NNE - north northeast; NSW - north southwest; NSE - north southeast; MED – Mediterranean; SNW - south northwest; SNE - south northeast; SSW - south southwest; SSE - south southeast (see Figure 1).

of the fleet categorical variable fleet (1 to 14) following a gaussian error distribution, centered at zero. As the size model is sex-specific, only fleets with sex information were included in the model.

Models were fitted using the function *bam* in *mgcv* package (Wood 2017) and parameters estimated by maximum likelihood method. Concurvity, a generalization of collinearity, was calculated to assess if a smooth term could be approximated by one or more of the other smooth terms using the function *concurvity* from *mgcv* package, if high concurvity was found one of the parameters would be excluded. Model selection was performed through an analysis of deviance. The significance of the explanatory variables (parametric and smooth terms) was assessed using function *anova.gam* in package *mgcv* (Wood 2017). If significant (F-test, $p<0.05$) the explanatory variable was retained in the model, otherwise it was excluded. All model variables and their interactions were significant (F test, $p<0.05$), except for the two-way interaction between Month and Sex. The two-way interaction was not excluded from the model, however, as the three-way interaction between the spatial effect with Month and Sex was considered significant. Additionally, no covariate was excluded due to concurvity. Therefore, the full model, containing all tested explanatory variables and interactions was used to explain the size distribution of swordfish.

Model validation was performed using several indicators. A residual analysis was carried out for the gamma model using package mgcViz (Fasiolo et al. 2020), and the deviance explained and r-squared calculated. To evaluate the predictive performance of the Gamma model, the root mean square error (RMSE) was calculated, with lower values indicating more precise predictions. The total deviance explained by the model was 28.6% with an adjusted r-squared of 0.28. The model had an RMSE of 28.2 (an average error of 28.2 cm between the predicted and the observed size). The residual analysis revealed no major trends or patterns in the residuals that could be considered problematic (Supplemental Material Figure S1). The partial effects of the smooth and parametric terms are presented in Supplemental Material (Figure S2).

Predictions of the spatial size distribution of swordfish and proportion of undersized fish (below) were obtained from the GAMM models using the *predict.gam* function of the *mgcv* package (Wood 2017), for a monthly grid of 1° × 1° with the associated environmental and physiographic data. For model predictions, SST data was obtained from the HadISST1 Met Office Hadley Center sea ice and sea surface temperature (HadISST1) data set (Rayner et al. 2003), using the *xtracto* function from the *rerddap* package (Chamberlain 2023) in R language (R Core Team 2023). The monthly predictions of size and probability of occurrence of undersized swordfish were averaged over each quarter to represent the seasonal variation. Thus, spatial maps were created by quarter. These plots are provided by sex for the size distribution model.

The expected mean size of swordfish varied spatio-temporally with different effects depending on sex. In general, in the South Atlantic Ocean the predicted mean size was larger than in the North Atlantic Ocean. The larger swordfish were predicted to occur at higher latitudes in both the North and South Atlantic Oceans, but also in equatorial, tropical and sub-tropical waters (Figure 5). The smaller specimens were predicted to occur mainly in the Gulf of Mexico and U.S. coast, and in the Eastern North Atlantic Ocean (Figure 5). An area with smaller specimens was also found in the Southeastern Atlantic Ocean. The distribution of smaller male swordfish was more extended in quarters 3 and 4 while larger sized males distribution is more expanded in quarters 1 and 2, occurring in the west North Atlantic both at high and low latitudes. For females, in quarters 3 and 4 larger individuals were less present in lower latitudes in the North Atlantic Ocean, while in quarters 1 and 2 the distribution of larger females was more extended in the western Atlantic Ocean. In the East Atlantic, both larger females and males seem to have a more expanded distribution in quarter 3. In the South Atlantic Ocean, larger mean sizes are predicted to be widely distributed while having a more contracted distribution in quarter 3 (Figure 5). Prediction plots of areas with few data points were interpreted with caution, as model fit for those areas may not be representative of the size distribution (see Figure S3 of supplemental material).

Distribution of undersized fish in the Atlantic Ocean

Spatio-temporal trends in proportion of undersized swordfish

Undersized fish were considered to be the fish below the corresponding minimum landing sizes implemented by ICCAT for swordfish. Currently in the Mediterranean Sea the minimum size is 100 cm LJFL (ICCAT, 2016) (the first implementation was of 90 cm LJFL (ICCAT, 2013)). For the North and South Atlantic Ocean there are two minimum landing size options that contracting parties can choose from for implementation, a 125 cm LJFL minimum size limit, which allows for a tolerance of fish below this size, or a 119 cm LJFL with no tolerance for fish below this size (ICCAT, 2017a, 2017b). For convenience, in the current work the minimum landing size was considered to be 125 cm LJFL for the Atlantic Ocean.

The proportion of undersized fish was calculated by area, quarter and year. The proportion of

Figure 5. Proportion of swordfish (*Xiphias gladius*) under 125 cm lower jaw fork length (LJFL) in the Atlantic Ocean (North and South) and 100 cm LJFL in the Mediterranean Sea by quarter from 1979 to 2022. CAN – waters offshore to Canada; GOM – Gulf of Mexico; NNW - north northwest; NNE - north northeast; NSW - north southwest; NSE - north southeast; MED – Mediterranean; SNW - south northwest; SNE - south northeast; SSW - south southwest; SSE - south southeast (see Figure 1).

undersized fish varied among quarters and areas (Figure 6). For the North Atlantic Ocean, all areas have percentages of undersized fish higher than 30%, with some areas reaching 60% in certain quarters. The exception was CAN where in the first quarter the percentage of undersized fish was relatively high (~40%) but it decreased, reaching a minimum of 10% in quarter 3. In the South Atlantic Ocean, percentages of undersized fish were lower than in the North Atlantic Ocean. The highest percentages (~30%) were found early in the year (quarter 1 and 2) in SNE. In the Mediterranean Sea, lower percentages of undersized fish occurred in quarters 2 and 3 while this percentage increased in quarters 4 and 1.

The time series trends in undersized fish seemed to be decreasing along the years for the various areas in the North Atlantic Ocean, while in the South Atlantic Ocean there was high variability in the yearly trends and in the Mediterranean Sea it was relatively stable (Figure 7).

Predicting proportion of undersized fish

A GAMM using a binomial distribution was constructed to predict the expected proportion of undersized fish, as detailed above for predicting size. Data on 450,954 individuals was used for modeling the proportion of undersized fish.

The GAMM model for proportion of undersized fish was expressed as follows:

$$\begin{aligned} g\left(P_{\text{undersized}}\right) = {} & \text{Month} + f\left(\text{Latitude: Longitude, Month}\right) \\ & + f\left(\text{Year}\right) + f\left(\text{SST}\right) + f\left(\text{Bathymetry}\right) \\ & + f\left(\text{dtl}\right) + \alpha\,\text{Fleet} \end{aligned}$$

The model used a binomial error distribution which is a discrete probability distribution of the number of successes (i.e. being undersized) in a sequence of n independent observations, and was fit with the logit link function (g). Definitions for the explanatory variables were the following:

Month as a categorical variable (January to December), f(Latitude:Longitude, Month) as a two dimensional smoother estimated with a spline on the sphere for the spatial effect by month, f(Year), f(SST), f(Bathymetry), f(dtl) as smoothers for the effect of the continuous variable year (1982-2022), sea surface temperature (SST), bathymetry and distance to coastline (dtl) estimated with a thin plate regression spline. α*Fleet* as a random intercept effect of the fleet categorical variable fleet (1 to 16) following a gaussian error distribution, centered at zero. Sex was not included as an explanatory variable as the minimum size regulation applies to both sexes.

Model fit, selection and validation was performed as detailed above, except for the predictive

Figure 6. Yearly proportion of swordfish (*Xiphias gladius*) under 125 cm lower jaw fork length (LJFL) in the Atlantic Ocean (North and South) and 100 cm LJFL in the Mediterranean Sea. Vertical lines indicate the introduction of the minimum landing size, starting in 1991 in the North and South Atlantic (note that for the South Atlantic the time series starts in 1996) and 2017 for the 100 cm LJFL in the Mediterranean Sea. CAN – waters offshore to Canada; GOM – Gulf of Mexico; NNW - north northwest; NNE - north northeast; NSW - north southwest; NSE - north southeast; MED – Mediterranean; SNW - south northwest; SNE - south northeast; SSW - south southwest; SSE - south southeast (see Figure 1).

performace, as RMSE does not apply to the binomial model, instead the probabilistic prediction accuracy was measured with the Brier score (Brier 1950), calculated using *BrierScore* function in package *DescTools* (Signorell 2023). This score ranges from 0 to 1, with lower values indicating greater accuracy. Additionally, the discriminative capacity of the Binomial model was determined by the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) curves (Fawcett 2006), with the calculation of the sensitivity (capacity to correctly detect the event, in this case defined as the capture of undersized fish) and specificity (capacity to correctly exclude the nonevents, in this case the capture of fish above the landing size limit). The AUC values range from 0 to 1, with a value of 0.5 indicating that the models do not have any discriminative capacity (i.e., indicating what would be obtained with a model with random performance), values between 0.7 and 0.9 indicating acceptable to excellent model discrimination, and values > 0.9 indicating that the models have outstanding discrimination (Hosmer and Lemeshow 2000). These calculations were performed using the *roc* and *coords* functions in package *pROC* (Robin et al. 2011). Partial effects were plotted using the *draw* function in package *gratia* (Simpson 2024).

All terms were considered significant (F-test, $p < 0.05$) and concurvity between terms was low. Therefore, the full model, including all tested explanatory variables and interactions, was used to explain the proportion of undersized swordfish. The total deviance explained by the model was 18.8% and had an adjusted r-squared of 0.21. The BS and AUC were 0.17 and 0.78, respectively, indicating a model with good accuracy and acceptable to excellent discriminative capacity. The partial effects of the smooth and parametric terms are presented in Supplemental Material (Figure S4).

Predictions were computed as described above and maps plotted by quarter. There were extensive areas where the predicted proportion of undersized fish varied between 0.2 and 0.4. Areas with a high proportion

Figure 7. Prediction of the mean size distribution of swordfish (*Xiphias gladius*) in the Atlantic Ocean by sex and quarter of the year from a Generalized Additive Mixed Model.

of undersized fish included the Caribbean Sea, off the North Brazilian coast, the Gulf of Mexico and U.S. coast, the waters around the Azores and Cape Verde and waters off the west coast of Africa (Figure 8). Areas with lower probabilities were mostly in the South Atlantic. Some seasonality occurred with higher proportions of undersized swordfish being more common in the North Atlantic Ocean in quarters 3 and 4, while in the South Atlantic Ocean there is a tendency for higher concentrations in quarters 4 to 2 in the east. Prediction plots of areas with few data points were interpreted with caution, as model fit for those areas may not be representative of the proportion of undersized fish (see Figure S5 of Supplemental material).

Figure 8. Prediction of the proportion of undersized (fish below 125 cm lower jaw fork length (LJFL) in the Atlantic stocks and 100 cm LJFL in the Mediterranean stock) swordfish (*Xiphias gladius*) in the Atlantic Ocean by quarter of the year from a Generalized Additive Mixed Model.

Discussion

Size and proportion of undersized spatio-temporal distribution

In ICCAT, the Mediterranean stock is managed independently from the Atlantic stocks. Genetic studies have shown that the Mediterranean specimens are genetically different from the Atlantic specimens (Viñas et al. 2007; Smith et al. 2015). Likewise studies using parasites also corroborate this separation (Mattiucci et al. 2014). Additionally, life history parameters such as growth and reproduction are also different, with Mediterranean swordfish reaching smaller maximum sizes and maturing earlier than Atlantic swordfish (Abid and Idrissi 2006). In this study, the maximum size of Mediterranean swordfish was also observed to be smaller than those in the Atlantic Ocean. Although maximum sizes in the North and South Atlantic Oceans are not reported to be different, the mean size for swordfish observed in this study was generally larger in the South Atlantic Ocean than in the North Atlantic Ocean. Differential growth between sexes has been reported for each of the stocks, with females growing slower and reaching larger sizes than males (Tserpes and Tsimenides 1995; Arocha et al. 2003; Millot et al. 2024). Additionally, previous studies of the sex ratio at size have shown increasing proportions of females in sizes above 150 cm, with females representing close to 100% of the catch at sizes larger than 200-215 cm (Mejuto 2018). The presented results are in agreement with these findings, as the larger individuals were mostly females and in all areas females showed larger mean size than males.

It is generally acknowledged that there is a differential distribution of swordfish by size (Palko et al. 1981; Hazin and Erzini 2008; Mejuto et al. 2008; Mejuto 2018). Mejuto et al. (2008) noted a greater catch of fish < 125 cm by Spanish longliners in the North Atlantic Ocean, while catches of fish >160 cm were greater in southern areas of the South Atlantic Ocean. In the Southwest Atlantic, Hazin and Erzini (2008) and Domingo et al. (2007) found juveniles (< 125 cm LJFL) to be concentrated in coastal waters. Hazin and Erzini (2008) also reported that maturing individuals (125-170 cm LJFL) were present in tropical waters and adults (>170 cm) in higher latitudes. Muhling et al. (2015) reported smaller fish being caught in the warmer waters of the Gulf of Mexico and U.S. east coast, while larger specimens were present in the Sargasso Sea and colder waters north of the Gulf stream. Results from the current study corroborate previous findings, showing larger individuals found at higher latitudes in both hemispheres, and the distribution of smaller specimens conformed to previous accounts depending on the season. The presence of larger individuals in colder water regions has been associated with increased feeding opportunities to sustain increased energy requirements and the capacity of larger fish to forage in colder waters (Neilson et al. 2013). Additionally, high-relief bottom structure and complex thermal structure seem to attract swordfish (Sedberry and Loefer 2001), for example where the warm Gulf Stream waters meet the colder Labrador current in the western North Atlantic Ocean; the warm Agulhas current meets the cold Benguela upwelling system in Southeast Atlantic Ocean; or the Brazil-Malvinas confluence in the Southwest Atlantic.

Swordfish movement has been described as complex and multidirectional (Palko et al. (1981). Different parts of the population may have different spatio-temporal distributions based on their needs given their life stages. North-south migrations, from temperate waters toward tropical areas in winter and back to temperate areas in spring/summer have been described for swordfish, and are mainly associated with large females (Neilson et al. 2013). Additionally, studies on sex ratio at size have shown that the proportions of females at size varies across areas in the Atlantic Ocean, suggesting differential movement of males and females into spawning or feeding areas (Mejuto et al. 2018). Latitudinal displacements related to movements between feeding and spawning grounds have been particularly well described through tagging for the western North Atlantic (Neilson et al. 2009; Neilson et al. 2014; Braun et al. 2019), but have also been reported for the central and eastern North Atlantic (Abascal et al. 2015) for maturing or mature individuals. For juveniles tagged in the Azores, the movements were more restricted, suggesting different foraging needs between immature and mature individuals and displacements changing to accommodate large scale movements related to spawning (Braun et al. 2019). In addition to the tagging information, U.S. fishery observer data showed a northward migration in the west Atlantic in boreal spring through boreal autumn, with a bias toward females (Schirripa et al. 2017). The quarterly predicted mean sizes also showed these movements, with larger fish having a more extended distribution toward tropical waters in quarters 1 and 2.

For the South Atlantic Ocean, limited information is available about movements across the basin from satellite tagging studies, although some swordfish have been tagged in particular areas (Matsumoto et al. 2003; West et al. 2012; Rosa et al. 2022). A limited number of conventional tags with recapture information show displacements between temperate and subtropical areas in the western South Atlantic (ICCAT 2023c). Chang et al. (2012) reported that habitat suitability changes seasonally, with preferred habitats in the July to October period more contracted toward the tropical area. Studies on catch rates in western south Atlantic waters indicate that latitudinal migrations also exist (Lima et al. 2023), showing increases in catch rates in equatorial waters in March and May, while catches increase in colder waters in southern areas in August and October. Domingo et al. (2007) found that the catch rates of larger specimens (>160 cm LJFL) in the southwest Atlantic Ocean increased in the third quarter in southern areas. Results also showed that larger mean sizes in the South Atlantic Ocean are more widely distributed from low to high latitudes, while in quarter 3 their distribution is more concentrated toward higher latitudes. Hazin and Erzini (2008) hypothesized that the presence of larger fish in southern waters may be related to increased food availability due to the occurrence of the subtropical convergence that increases primary production. Movements to temperate waters in the austral winter appear to be contrary to what is reported in the North Atlantic Ocean. In the Southeast Atlantic although information is scarce, larger sized swordfish are reported to occur in South African catches (Govender et al. 2016), as predicted by the size distribution model. A portion of swordfish in this area may originate from the Indian Ocean stock, as there is evidence that the area off the Cape of Good Hope is a common feeding area for both the South Atlantic and the Indian Ocean stocks (West 2016). More studies, possibly using satellite tagging, are necessary to better understand large scale horizontal movements in the South Atlantic Ocean.

In the North Atlantic Ocean, spawning begins in the northwest in December in oceanic areas southwest of the Sargasso Sea, at lower latitudes, and progresses to higher latitudes with the expansion of the 24 °C isotherm to areas as far North as 35°N (Arocha 2007). The size distribution within the GOM area corroborates this as a nursery area for swordfish, given the strong bi-modal distribution, with the first peak occurring close to 90 cm. Given the rapid growth of swordfish in the early years, fish of this size are probably young of the year (Arocha et al. 2003). A preference for warmer temperatures has been reported for early juvenile swordfish, which are considered to be less migratory and remain in the areas close to spawning or follow the 24 °C isotherm (Mejuto 2003). Thermal preferences and the westward drift due to predominant currents can favor the abundance of larvae and smaller juveniles in the west Atlantic, while transport may occur to the central Atlantic through the Gulf Stream current (Govoni et al., 2003). High proportions of undersized fish are predicted to occur in waters closer to shore, either continental or associated with islands and archipelagos. Most of the U.S. coast, the Gulf of Mexico and Caribbean Sea and the east North Atlantic, including the Cape Verde, Canary, Madeira and Azores islands have an estimated high probability of occurrence of undersized fish. Hoey and Mejuto (1991) also report high proportions of undersized fish occurrence in the west Atlantic, extending across the tropical Atlantic toward the African coast, with presence in the northeast Atlantic but with less predominance. The proportion of undersized fish increased particularly in quarters 3 and 4 for most areas, being consistent with that reported by Hoey and Mejuto (1991), which suggested this could be an indication of fall recruitment of later winter early spring spawning. In the northeast Atlantic Ocean, increased catches of swordfish occur in autumn and winter (Coelho et al. 2022b), Parra et al. (2023) suggested this increase could be related to increased feeding opportunities due to upwelling regimes taking place earlier in the year. Undersized fish presence may also benefit from the increase of prey availability in these areas and quarters. Given the genetic evidence that swordfish from the Mediterranean are found in northeast Atlantic waters (Smith et al. 2015) and as swordfish in the Mediterranean Sea have a smaller size, it cannot be excluded that a portion of the fish in this area are from the Mediterranean Sea, which could influence the observed sizes and proportion of undersized fish.

In the South Atlantic Ocean, spawning is thought to occur throughout the year in equatorial waters and at higher latitudes from January to June (Mejuto and García-Cortés 2014). In general, low proportions of undersized fish occur in this area and the occurrence of smaller individuals is mostly concentrated in coastal areas. A preference for coastal waters by juvenile fish has also been reported for the eastern Mediterranean Sea (Damalas and Megalofonou 2014). Spawning in the Mediterranean Sea occurs in well-defined areas from June to August, when the 24 °C isotherm occurs in this basin (Arocha 2007). Recruitment to the fishery of young of the year in the Mediterranean Sea starts between late summer and early autumn, and continues through winter (Tserpes and Peristeraki, 2007; Damalas and Megalofonou 2014). The increased proportion of undersized fish in the Mediterranean Sea observed for quarters 4 and 1 corroborates those findings.

Size regulation and management implications

Fisheries management can apply several policies or technical measures to meet management objectives. The measures most commonly used include catch limits, minimum landing sizes, gear restrictions, and time and/or area closures (Liu et al. 2016). In order to achieve the objectives swiftly, in many cases, several of these measures are applied simultaneously. In ICCAT, management recommendations to rebuild the swordfish stocks have been in place since 1991 for the Atlantic stocks, and since 2013 for the Mediterranean stock (ICCAT, 2016, 2017a, 2017b). One of the measures implemented is a minimum size limit, which has been in place for the Atlantic stocks since 1991. It was then revised in 1996, which makes it one of the oldest regulations in ICCAT still active. Regarding this measure, the objective of protecting small swordfish is stated in each of the stock management recommendations (ICCAT, 2016, 2017a, 2017b). Despite the long application of the minimum size regulation, its effectiveness as a protection for juvenile fish has been questioned due to high swordfish hooking mortality (Coelho and Muñoz-Lechuga 2019). For undersized fish captured in longline fisheries the hooking mortality can range from 63.3% to 88% (ICCAT, 2017c; Schirripa 2022). It has been noted that all sources of discard mortality (including hooking and post-release mortality) when applying minimum landing sizes can lead to reductions in maximum yield due to growth overfishing and loss of biomass to discard mortality (Coggins et al. 2007).

Under a minimum size limit regulation, it is hypothesized that the fleets could either avoid areas of high juvenile concentrations or modify the fishing gear to reduce incidental catches of undersized fish, which in both cases would result in lower discards and discard mortality of undersized fish (Schirripa 2022). Cramer (1996) reported for the U.S. pelagic longline fleet that the minimum size regulation had limited effects as the number of swordfish discarded dead had increased and the fleet did not seem to have changed its fishing patterns close to the US coastline. In general, for the North Atlantic Ocean, the proportion of undersized fish has been decreasing. A decreasing trend in discards was also estimated in the latest North Atlantic stock assessment (Schirripa 2022), however, it was noted that the reason for this decreasing trend could be either because the encounter rate of undersized fish had decreased or there was less recruitment.

Apostolaki (2005) noted that the effectiveness of the minimum size would depend, amongst other factors, on the discard survival of swordfish and could even increase the effort as more legal sized fish would need to be caught to offset what is discarded to reach the quota resulting in even more discards. In the Mediterranean Sea, Ortiz (2020) estimates that the increase in minimum size from 90 cm LJFL to 100 cm LJFL has led to an increase in discards, and García-Barcelona et al. (2020) reported for the Spanish fleet operating in the Mediterranean Sea an increase in effort after the implementation of the 100 cm LJFL minimum size regulation. For the North Atlantic stock, Schirripa (2022) also found that increasing the minimum size under high post-release mortality rates (including hooking mortality) would not influence the spawning potential ratio and this should be a function of effort, noting that in these cases management measures that decrease the encounter rate of undersized fish would be more effective. Assessing the effectiveness of the minimum size limit measures, as requested by ICCAT (ICCAT 2017a, 2017b) is complex, and even more challenging when little data on discards and/or catch-at-size information is available as well as potential changes in fleet distribution to avoid undersized fish. North Atlantic swordfish is currently undergoing a Management Strategy Evaluation (MSE) process, in which testing the effectiveness of the minimum size was set as an objective since the onset and has recently been identified as a priority robustness test (ICCAT, 2023c).

Simulation studies in the Mediterranean Sea showed that a spatial closure in the recruitment period to the fishery could result in long term increases in spawning stock biomass (Tserpes et al. 2009). Damalas and Megalofonou (2014) suggested that given the juvenile preference for coastal waters, while adults are more common in open waters, a combination of seasonal and spatial closure in coastal areas could help protect the immature fraction of the population. At the ICCAT level, closed fishing seasons have been in place for Mediterranean swordfish since 2008 (ICCAT, 2020). In the North Atlantic Ocean, time/area closures are also in place at national levels in 5 areas in the U.S. (NOAA, 2019) to protect juvenile swordfish. Apostolaki (2005) noted that a considerable proportion of the young of the year would need to be protected by these areas to have a positive effect on the population. Applying fishery closures to protect small sized swordfish in the Atlantic Ocean would potentially mainly benefit the North Atlantic stock given that in the South Atlantic Ocean there has been a lower proportion of undersized fish reported. Given the protracted spawning period and the wide distribution of fish over 125 cm LJFL in the North Atlantic, and the poorly known spawning, nursery, and migrations routes in the South Atlantic Ocean, the implementation of fishery closures for these stocks could be difficult.

Boerder et al. (2019) argued that marine protected areas or fishery closures when well designed, implemented, and in complementarity with other measures could be beneficial to highly migratory species. Relano and Pauly (2022) suggest using the philopatric behavior of highly migratory species to protect key migratory routes and stocks with the creation of “blue corridors”. The implementation of time-area closures, or dynamic fishery closures (Pons et al. 2022) could benefit the complete demographic structure of swordfish, by protecting both young and adult specimens. Poisson and Fauvel (2009) suggested that larger females might contribute more to the stock than younger females, and protecting these specimens could preserve the population age structure and lead to higher productivity. Alternatively, demographic models for swordfish in the Pacific Ocean have shown that age at first capture should mainly ensure the protection of the first 1-2 mature age classes (Au 1998), therefore there is still a need to fully understand what ages or sizes represent the higher reproductive value for the swordfish populations. For the implementation and assessment of the effectiveness of spatial protection measures a good understanding of the life history and population structure, migration routes, and fishing effort distribution is needed (Coelho and Muñoz-Lechuga 2019; Boerder et al. 2019; Relano and Pauly 2022).

In terms of management advice, the implementation of measures additional to the minimum retention sizes, such as best handling practices for the release of the undersized specimens might to some extent improve survivorship, however with the reported levels of at-haulback mortality noted above might not make such types of measures truly efficient in diminishing fishing mortality. On the other hand, alternative measures such as spatial and/or seasonal restrictions to protect particular life stages are possible. Due to the very wide Atlantic areas where the undersized fish are present, as shown in the present work, the implementation of such types of restrictions in such wide scale areas might be complex. The protection of larger individuals would represent smaller areas of the Atlantic, but as mentioned before there is still the need to fully understand what ages/sizes have the highest reproductive value for swordfish, so that it is clearer what areas should be prioritized in terms of future protection. Simulation frameworks (e.g. MSE) coupling biological knowledge, as provided in the current study, with fishery data and management goals might become essential in the implementation of management measures that will effectively contribute to swordfish recovery and conservation.

Data and modeling limitations

As fishery independent data sources are generally lacking for oceanic species, there is usually high reliance on fishery dependent data for these species. Data from commercial fleets can provide useful insights into spatio-temporal distribution of tuna and tuna-like species (Mourato et al. 2010), however, analysis of these data may have some caveats. Mainly, the size ranges and abundance reported by each fleet are affected by the species spatio-temporal availability and gear selectivity of each fleet (Fernandez-Carvalho et al. 2015; Coelho et al. 2018), which can influence the interpretation of results. These considerations are also important in this study which includes data from several fleets operating in different spatio-temporal strata, using different fishing gear and targeting several species. The quality of the data might also vary by fleet, and the observer coverage is low in general. Spatial sample availability depends on the spatial fleet distribution - in the current work the major area gaps are in the central North Atlantic and a portion of the Caribbean basin. These areas also have low reported catches (ICCAT, 2023c). Interpretation of results for areas with low sampling requires the recognition of this caveat.

The observed data of this study represents the portion of swordfish at a location that was selected by fishing operators with different fishing gears, and may not be representative of the whole size classes that were available in the wild (gear selectivity effect). It should be noted that most of the data used in this study are from pelagic longlines, targeting mainly swordfish, sharks or tunas, although harpoon and gillnet data were also included. While gillnets are likely less selective; the harpoon fishery is more selective, catching larger swordfish (Hanke and Gillespie 2022), basking at the surface during the daytime (Neilson et al. 2013). In the case of pelagic longlines, selectivity may depend on various factors, including gear depth, hook size and type, bait and leader type and soaking time. Based on recent meta-analysis, hook type has been shown to influence the retention of swordfish; however, neither bait nor leader type seem to influence retention (Gilman et al. 2020; Santos et al. 2023). Nonetheless, individual studies for particular areas have shown that both bait and leader can have an influence (Santos et al. 2012; Forselledo et al. 2014; Amorim et al. 2015; Fernandez-Carvalho et al. 2015; Santos et al. 2024). Chancollon et al. (2006) reported on the positive relationship between prey and swordfish size, noting however that previous studies did not find this positive relationship. Swordfish have an opportunistic generalist diet, that is mostly composed of cephalopods and teleost fish prey, with diets varying regionally (Logan et al. 2021).

Environmental variables have also been associated to changes in swordfish distribution and preferred habitats (Hazin and Erzini 2008; Chang et al. 2012; Change 2013; Lima et al. 2023). In this study, bathymetry, distance to land and sea surface temperature were included in the modeling and found to be significant, both to explain the mean size as well as the proportion of undersized fish. Similarly, Hazin and Erzini (2008) also found environmental variables to have different effects in the distribution of immature, maturing and mature swordfish. Given the vertical habitat use of swordfish, recent studies on swordfish catchability have shown that including subsurface environmental parameters increased model performance, while noting that SST significantly contributed to explain catchability (Tang et al. 2024). Further consideration of the effects of environmental variables, particularly at depth, in the size distribution of swordfish should be included in future studies. Model predictive skill may be affected by all the caveats presented.

Conclusions

This study provides the most comprehensive Atlantic Ocean wide size distribution analysis of swordfish to date and makes important contributions to the understanding of the species spatiotemporal and seasonal variations. The records used include data from scientific fishery observer programmes, electronic logbooks, port-sampling and research cruises, which makes this the most complete and comprehensive dataset available and analysis carried out at an Atlantic oceanic-wide level.

The main results support the previously described distribution of swordfish in the Atlantic Ocean, with the most extreme records at 56°N and 45°S. Nakamura (1985) had previously reported swordfish to be the most widely distributed billfish species, occurring from 60°N to 50°S in the Atlantic Ocean, which is in line with the results now presented.

In terms of sizes, the mean predicted swordfish sizes in the South Atlantic Ocean tended to be larger than in the North Atlantic Ocean, and overall, the larger swordfish tended to occur at higher latitudes of both hemispheres. In terms of proportions of undersized swordfish, there were extensive areas along the Atlantic Ocean where proportions varied between 0.2 and 0.4, areas with highest proportion of undersized fish included the Caribbean Sea, off the North Brazilian coast, the Gulf of Mexico and U.S. coast, the waters around the Azores and Cape Verde and waters off the west coast of Africa. There also seemed to exist an association of smaller and undersized swordfish with coastal waters. The size distribution differed between males and females, and also varied by quarter, indicating movements of large fish between temperate and tropical waters.

The size-specific distribution patterns presented provide a better understanding of different aspects of the swordfish distribution in the Atlantic Ocean, which can be used to enhance scientific advice for managing this species. While this work provides the first Atlantic-wide scale work on the distribution aspects of the species, there might be the need for specific and more detailed analysis in specific areas not fully covered in this work. It is expected that this and further complementary analyses can contribute to a better knowledge of the sex-specific spatial-seasonal dynamics of the species in the Atlantic Ocean, particularly its dynamics at size by sex, to further improve scientific advice and ultimately fisheries management.

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